2021 10th International Conference on Internet of Everything, Microwave Engineering, Communication and Networks (IEMECON) | 978-1-6654-2686-2/21/$31.00 ©2021 IEEE | DOI: 10.1109/IEMECON53809.2021.9689172

Image Splicing Detection and Localisation using EfficientNet and Modified U-Net Architecture

Md Shamim Ahmed
Department of Information Technology
Indian Institute of Engineering Science and Technology
Shibpur, INDIA - 711103
shamim.pg2019@it.iiests.ac.in

Dr. Ruchira Naskar
Department of Information Technology
Indian Institute of Engineering Science and Technology
Shibpur, INDIA - 711103
ruchira@it.iiests.ac.in

***Abstract*—The process of preserving, identifying, extracting, and documenting computer evidence that can be utilized in a court of law is known as digital forensics. It is the science of finding evidence from digital media like a computer, mobile phone, server, or network. It equips the forensic team with the most up-to-date techniques and tools for resolving complex digital cases.. Among many of them, the major problem which is addressed in this project is Image Forgery Detection. More precisely we deal here with the "splicing attack detection and localization". Even though there are numerous techniques to conquer the problem, still there are loopholes in them like the accuracy of the forged region localization. As a part of this project, we propose a Convolutional Neural Network (CNN) based Deep Learning architecture, which is capable of detecting this forgery with state-of-the-art performance. We propose a transfer learning (EfficientNet) based image classification algorithm for the classification of forged Images and EfficientNet based U-Net Architecture for localization. In the classification part, we extract the image patches from the forged regions and non-forged regions in an image and classify them. The classification fetched us state of the art result. In the localization part, we have built a U-Net architecture by replacing the U-Net Encoder part with EfficientNet layers and keeping the decoder part as usual. We then pass the forged images through the network to train the model. The localization experiment also fetched state-of-the-art results.**

***Keywords*—Digital Forensics; Image Forgery Detection; Splicing Attack; Convolutional Neural Network; Instance Segmentation; U-Net; Transfer Learning;**

I. Introduction

Recently, we have witnessed advances in user-friendly image editing software, image editing techniques. So image manipulation has become very easy and affordable [1]. Forgers may employ forged photos to transmit misleading information, spread rumours, or give false testimony, all of which have detrimental social consequences. Image forensics, which aims to differentiate counterfeit photographs and prevent forgers from utilising them for unethical business or political reasons, has sparked a lot of interest in the scientific and industrial areas. To find the altered regions, various clues are analysed like the splicing, copy-move, and removal, etc. The image splicing basically copies regions from an authentic image and pastes them to other images, copy-move forgery copies and pastes regions within the same image. It's really hard to identify the tampered regions even after careful inspection. Examples of these manipulations are shown in Fig. 1.

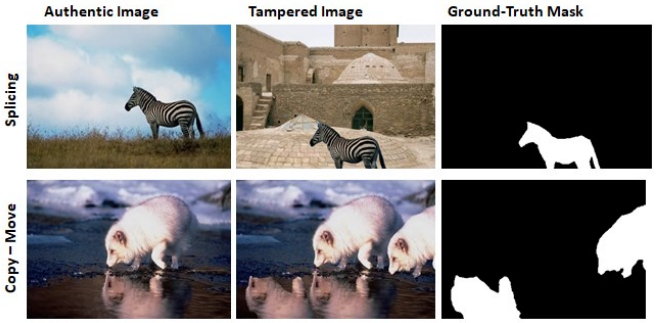


Fig. 1. Examples of tampered images which are result of different tampering techniques. These are the examples of Splicing and Copy-Move forgery attacks.

The traditional feature extraction-based methods can be categorised into four types: detection methods based on the essence attribute [2] [3] [4], detection methods based on the imaging device attribute [5] [6] [7] [8], detection methods based on the image compression attribute [9] [10] [11], and detection methods based on the hash techniques [12] [13] [14]. However, the traditional features extraction techniques persist with lots of drawbacks. i) Firstly image essence techniques fail if some hidden processes are performed after the process of splicing forgery. ii) In case the device noise intensity of an image is weak, the detection methods based on the imaging device fails. iii) The detection based on the image compression attribute only detects the images saved by JPEG format. iv) The hash techniques rely on the hash of the original untampered image. So we cannot rely on one single forgery detection method.

Convolutional neural networks (CNN) have achieved great success in the research area of computer vision. CNN has been used by researchers to accomplish splicing forgery detection. CNN is originally used to judge whether the image has been forged in [15], but it couldn't locate the tampered regions. In [15] the authors tried to locate the tampered regions by CNNs, but the detected regions showed inaccurate rough areas that are made up of some square white blocks. To improve researchers tried to improve the detected tampered regions, the methods [16] [17] use the non-overlapping image patch as the input of CNNs. However, when an image patch totally comes from the

978-1-6654-2686-2/21/$31.00 ©2021 IEEE

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 06:59:52 UTC from IEEE Xplore. Restrictions apply.

tampered regions, this image patch will be judged un-tampered label. In [18], the authors use a larger picture patch to disclose the image properties of the tampered portions; nevertheless, if the forgery image is small, the detection method may fail. For the existing CNN-based detection methods, since they use the image patch as the input of the network, the contextual spatial information is lost, which easily causes incorrect prediction. Furthermore, when the network architecture becomes more complex, the gradient degradation problem emerges, and feature discrimination weakens, making splicing forgery detection increasingly difficult, if not impossible.

Recent work on image forensics witnessed good advances. In [19] authors identified tampering of the images using local descriptors based on variations in visual elements like texture, illumination, or color in images. However, it's not guaranteed to detect forged regions using visual elements, the model didn't promise a good performance. In [20] the authors used two networks, one (C-CNN) to detect forged regions and the other (R-CNN) to filter the inaccurately detected forged regions. Both are traditional methods to extract the image features. The models couldn't achieve appreciable results.

II. Proposed Methods

A. Proposed Splicing Detection Model

To improve the classification accuracy and optimum localization of the forged regions, we propose two novel Image classification and localization networks. In the classification part, we transformed the images in such a way so that our algorithm could learn the features more precisely and fetch us optimum results. For the initial part, we took the idea from this [15] paper. Every fictitious image has a mask that corresponds to it. We utilize that mask to sample the forged image along the spliced region's boundary in such a way that both the forged and unforged parts of the image contribute at least 25% of the total. These samples will have distinct boundaries that are only present in bogus photographs. The CNN we develop will have to learn these boundaries. Similarly, we extract a similar number of patches from the authentic images. We then classify them using our CNN model. We used transfer learning, EfficientNet [21] for the classification purpose. EfficientNet is the recent state-of-the-art ImageNet competition classification model. We achieved state-of-the-art results after performing the experiment.

Image classification of this experiment is about classification between forged and authentic images. We have chosen the CASIA 1 dataset for this experiment. The dataset consists of authentic images, spliced images, and masks related to the forged images. Splicing attack forgery detection takes place when we paste a portion of an image on top of another image. As we have discussed above, we try to extract image patches of the boundaries from the forged image which consists of pixels from the forged portion as well as pixels from an authentic portion of an image. We identify the boundaries using the mask of each image. The mask contains a white portion (255) which is the forged and a black portion (0) which is authentic. Programmatically we extract boundary image

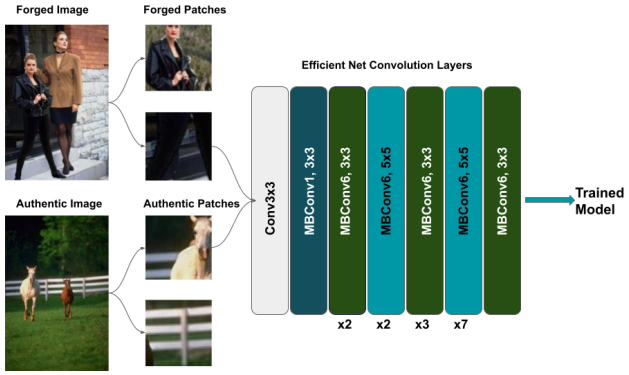


Fig. 2. Flow diagram of classification of forged and authentic images.

patches that contain both these types of pixels. Similarly, we extract a similar number of patches from authentic images. These boundaries are to be learned by the CNN architecture that we design.

To make boundaries even more distinct, the gray-scale images were converted to binary Otsu's thresholding (implemented in Open CV) after de-noising using the Gaussian filter. After this operation, sampling was merely a matter of moving 64*64 windows (with a stride of 8) through the fake image and counting 0 valued(black) pixels in the corresponding mask and sampling in case the value lies in a certain interval.

B. Proposed Splicing Localisation Model

In the case of the localization part, we use transfer learning-enabled U-Net architecture [22]. The U-Net architecture consists of encoding and decoding layers. In the encoding part, we used layers of EfficientNet and for decoding, we used the default layers of U-Net. This experiment also fetched us state-of-the-art results. The inference part of the forged images was very impressive performing above all the existing forgery localization algorithms.

We know that Image classification using transfer learning architectures consists of multiple layers of neural networks. There is a strong chance of overfitting in such cases. To avoid the same, there is a requirement for a large dataset. The CASIA1 dataset contains 462 spliced images which are solely not sufficient to train the best model. Therefore we have combined both CASIA1 and CASIA2 datasets containing 462 and 1765 images respectively. Now 2226 spliced and authentic images individually are good numbers to train our model.

III. Experimental Results

A. Dataset

In order to extract the image patches for classification of the forged images, we have considered CASIA [23] dataset. It contains 1,275 sets of images (1,275 forged images and 1,275 original images along with their corresponding masks) in JPG and PNG formats. Most of them have a pixel resolution of approximately 384 * 256. In the case of the second experiment, we did use deep neural networks like RESNET and EffcientNet. Due to this reason, we required an abundance

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 06:59:52 UTC from IEEE Xplore. Restrictions apply.

of images to get the best possible model for the segmentation task. We combined both the CASIA 1 and CASIA 2 datasets. Both the datasets combined 2226 images of nearly the same configurations as CASIA 1 dataset.

B. Evaluation Metrics

The number of correctly detected tampered pixels (TP), the number of wrongly detected tampered pixels (FP), and the number of incorrectly detected un-tampered pixels (FN) are important evaluation metrics for detecting an image splicing forgery. The Precision, Recall, and F-measure are utilised to evaluate the performance of the suggested splicing forgery detection algorithms in the following tests.

Here, Precision is defined in Eq.(1), Precision is the probability that the detected regions in the ground-truth image are truly tampered regions, Recall is the probability that the tampered regions in the ground-truth image will be detected as defined in Eq.(2), and the F1 measure synthesises Precision and Recall into a single measurement to synthetically evaluate detection performance., as defined in Eq.(3).

$$Precision = \frac{TP}{TP + FP} \tag{1}$$

$$Recall = \frac{TP}{TP + FN} \tag{2}$$

$$F - measure = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{3}$$

C. Deep Learning Model using Transfer Learning for classification

We leveraged transfer learning using the pre-trained models on the ImageNet dataset for classification purposes. Initially, we used the ResNet50 model with 'binary cross-entropy loss and Adam optimizer having a learning rate of 1e-6. We have run the model for 15 epochs. It fetched a test accuracy of 94.15% with a loss of 0.31.

To improve further we leveraged the latest state-of-the-art ImageNet model, EfficientNet. AutoML MNAS created the EfficientNet-B0 baseline network, and Efficient-B1 to B7 are acquired by scaling up the baseline network. The architecture uses the mobile inverted bottleneck convolution (MBConv). The researchers then scaled up this baseline network to create the EfficientNets family of deep learning models. Firstly, we used the EfficientNetB0 as the base architecture with Adam optimizer (learning rate of 1e-3). It fetched an accuracy of 97.12% with a 0.11 loss. The steps are depicted in Fig. 2.

To further increase the accuracy we leveraged the EfficientNetB2 with Adam optimizer (learning rate of 1e-3). It achieved an accuracy of 98.26% with a loss of 0.054. The result was very close to state-of-the-art accuracy. To further improve the accuracy we leveraged the EfficientNetB3. We use the Adam optimizer (learning rate 1e-4). Finally, we achieved the state of the art accuracy of 99.02% with a loss of 0.046.

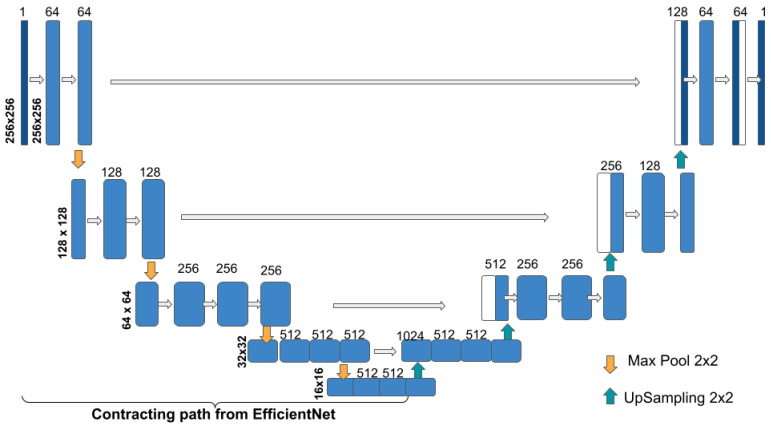


Fig. 3. The proposed U-Net Architecture with Efficient-Net Encoding.

D. U-Net Architecture using Transfer Learning for Localisation

As discussed above we have used U-Net Architecture for the localization of the spliced regions. U-Net is a state-of-the-art Image segmentation algorithm. We have incorporated some major changes in the architecture to fetch out optimum accuracy. The U-Net architecture consists of two parts encoding and decoding. Since the layers of the architecture can be easily modified or replaced as per our requirement, we have leveraged transfer learning layers in the encoding part. The EfficientNet architecture is the modern state-of-the-art Image classification architecture that fetched optimum complexity with minimum use of deep networks. We have changed the encoding part with the EfficientNet layers with a similar number of neurons in each layer. The decoding part of the U-Net is left as it is. Apart from the Precision, recall, and f1-score. We would be using dice-coefficient here, which is essentially used for semantic segmentation performance analysis.

We have firstly incorporated the EfficientNetB0 as the encoding layer of our network. Similarly, we have extracted few skip connection layers and concatenated them with the encoder output to perform the upsampling operation. We have used relu activation function and batch normalization at each subsequent layer. The architecture is a semantic segmentation technique that essentially links pixels to the class label. The class label here is forged and un-forged regions. The regions and their corresponding pixels are identified by the model using the ground truth of each spliced image. We can think of semantic segmentation as image classification at a pixel level. The performance of the EfficientNetB0 wasn't up to the mark. We repeated the same procedure with other EfficientNet Architectures as well. Subsequently, we incorporated the EfficientNetB7 as our encoding layer and that fetched us the state-of-the-art result. The model secured a dice coefficient of 0.78, precision of 0.86, recall of 0.74, F1-score of 0.8, and AUC of 0.87.

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 06:59:52 UTC from IEEE Xplore. Restrictions apply.

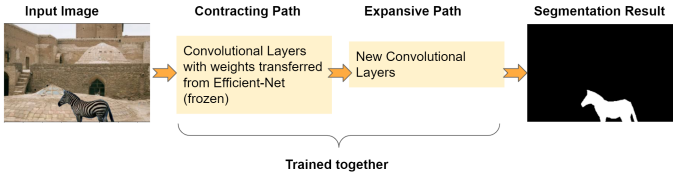


Fig. 4. Steps for the localisation.

E. Inferences

Above are the inferences of our results. We got nearly accurate output of the bounding/forged regions. It's evident from our performances of the resulting output. Here the first column corresponds to the spliced images, middle corresponds to the mask/ground-truth of the forged regions and the third column represents the resulting output.

The following are the comparisons of our experimental results with various other techniques which are been incorporated before using CASIA dataset. Few of the spaces are kept blank because the papers didn't use that concerned metric and hence we couldn't get the data. So '-' denotes that the result is not available in the literature.

TABLE I
CLASSIFICATION PERFORMANCE ANALYSIS OF RECENT MODELS

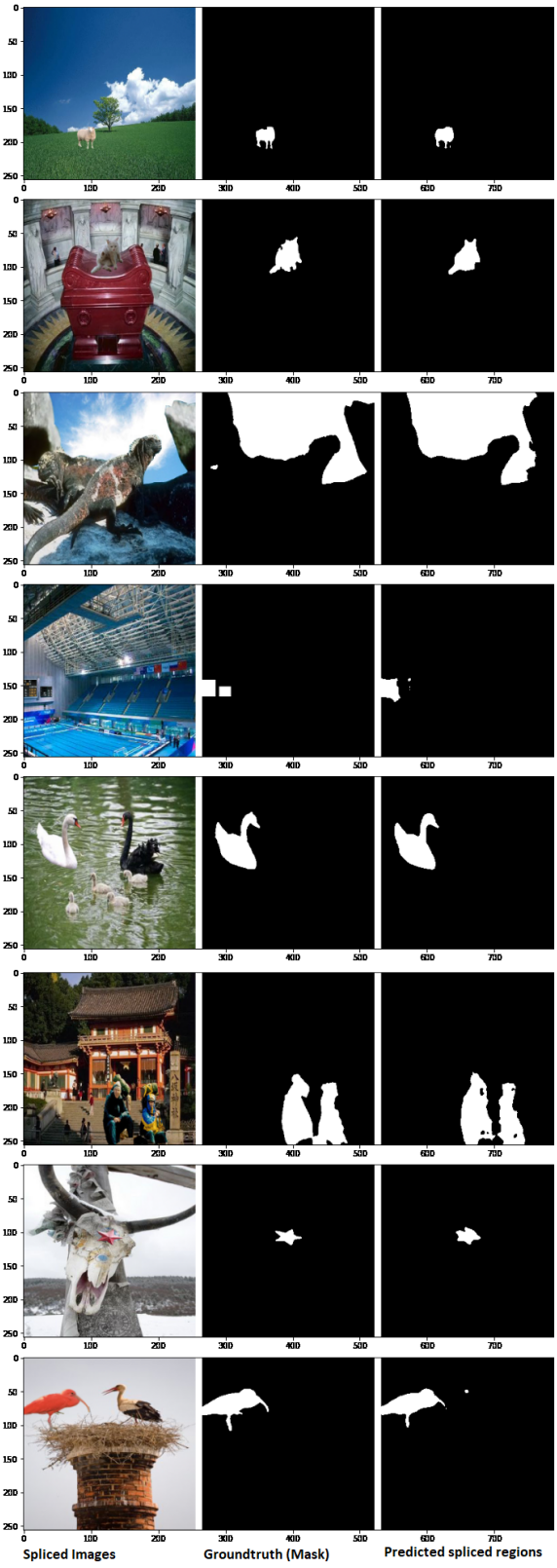
	Precision	Recall	F1-Score	Acc.
Illumination-Reflectance [24]	97	99	98	94.5
Low-dim DCT and DWT [25]	-	-	-	96
Wavelet transform [26]	98	97.08	97.53	95.45
Optimal threshold [27]	-	-	-	98.25
Local descriptor [19]	-	-	-	96.97
CFA Artifacts [28]	99	99	99	98.15
Our Model	99	99	99	99.02

TABLE II
LOCALISATION PERFORMANCE ANALYSIS OF RECENT MODELS

	Precision	Recall	F1-Score	AUC
Error Level Analysis [29]	-	0.214	0.613	-
Noise Inconsistency [30]	-	0.263	0.612	-
Coarse to Refined CNN [17]	0.581	0.808	0.6758	-
Local Descriptor [19]	-	-	0.58	-
Rich Features[31]	-	-	0.408	0.795
Our Model	0.866	0.7471	0.805	0.8790

IV. CONCLUSION

In this work, Image Splicing Detection and Localisation using EfficientNet and Modified U-Net Architecture is proposed for splicing attack forgery detection and localisation. In classification part we have extracted the image patches from the forged and authentic regions to distinguish the border parts better and classify using transfer learning technique between the forged image and the authentic image. In case of localisation we have used a modified U-Net architecture embedding the EfficientNet architecture in the encoder part and conjunction the same with the decoding part. Both the classification and the localisation part fetched us state of the art results. The work is capable for usage in practical applications. However there is still scope of improvement in the image localisation part.



Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 06:59:52 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] Y. Wu, W. Abd-Almageed, and P. Natarajan, "Busternet: Detecting copy-move image forgery with source/target localization," in *Proceedings of the European Conference on Computer Vision (ECCV)*, September 2018.

[2] W. Chen, Y. Q. Shi, and W. Su, "Image splicing detection using 2D phase congruency and statistical moments of characteristic function," in *Security, Steganography, and Watermarking of Multimedia Contents IX*, E. J. D. III and P. W. Wong, Eds., vol. 6505, International Society for Optics and Photonics. SPIE, 2007, pp. 281 – 288. [Online]. Available: https://doi.org/10.1117/12.704321

[3] Wei Wang, J. Dong, and T. Tan, "Effective image splicing detection based on image chroma," in *2009 16th IEEE International Conference on Image Processing (ICIP)*, Nov 2009, pp. 1257–1260.

[4] X. Zhao, J. Li, S. Li, and S. Wang, "Detecting digital image splicing in chroma spaces," in *Digital Watermarking*, H.-J. Kim, Y. Q. Shi, and M. Barni, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2011, pp. 12–22.

[5] H. Gou, A. Swaminathan, and M. Wu, "Noise features for image tampering detection and steganalysis," in *2007 IEEE International Conference on Image Processing*, vol. 6, 2007, pp. VI – 97–VI – 100.

[6] Y. Hsu and S. Chang, "Detecting image splicing using geometry invariants and camera characteristics consistency," in *2006 IEEE International Conference on Multimedia and Expo*, July 2006, pp. 549–552.

[7] M. K. Johnson and H. Farid, "Exposing digital forgeries through specular highlights on the eye," in *Information Hiding*, T. Furon, F. Cayre, G. Doërr, and P. Bas, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2007, pp. 311–325.

[8] B. Mahdian and S. Saic, "Detection of resampling supplemented with noise inconsistencies analysis for image forensics," in *2008 International Conference on Computational Sciences and Its Applications*, June 2008, pp. 546–556.

[9] M. K. Johnson and H. Farid, "Exposing digital forgeries in complex lighting environments," *IEEE Transactions on Information Forensics and Security*, vol. 2, no. 3, pp. 450–461, Sep. 2007.

[10] Z. Lin, J. He, X. Tang, and C.-K. Tang, "Fast, automatic and fine-grained tampered jpeg image detection via dct coefficient analysis," *Pattern Recognition*, vol. 42, no. 11, pp. 2492 – 2501, 2009. [Online]. Available: http://www.sciencedirect.com/science/article/pii/S0031320309001198

[11] S. Ye, Q. Sun, and E. Chang, "Detecting digital image forgeries by measuring inconsistencies of blocking artifact," in *2007 IEEE International Conference on Multimedia and Expo*, July 2007, pp. 12–15.

[12] Z. Tang, X. Zhang, X. Li, and S. Zhang, "Robust image hashing with ring partition and invariant vector distance," *IEEE Transactions on Information Forensics and Security*, vol. 11, no. 1, pp. 200–214, Jan 2016.

[13] X. Wang, K. Pang, X. Zhou, Y. Zhou, L. Li, and J. Xue, "A visual model-based perceptual image hash for content authentication," *IEEE Transactions on Information Forensics and Security*, vol. 10, no. 7, pp. 1336–1349, July 2015.

[14] C. Yan, C. Pun, and X. Yuan, "Quaternion-based image hashing for adaptive tampering localization," *IEEE Transactions on Information Forensics and Security*, vol. 11, no. 12, pp. 2664–2677, Dec 2016.

[15] Y. Rao and J. Ni, "A deep learning approach to detection of splicing and copy-move forgeries in images," in *2016 IEEE International Workshop on Information Forensics and Security (WIFS)*, Dec 2016, pp. 1–6.

[16] J. H. Bappy, A. K. Roy-Chowdhury, J. Bunk, L. Nataraj, and B. S. Manjunath, "Exploiting spatial structure for localizing manipulated image regions," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, Oct 2017.

[17] Y. Wei, X. Bi, and B. Xiao, "C2r net: The coarse to refined network for image forgery detection," in *2018 17th IEEE International Conference On Trust, Security And Privacy In Computing And Communications/ 12th IEEE International Conference On Big Data Science And Engineering (TrustCom/BigDataSE)*, Aug 2018, pp. 1656–1659.

[18] B. Liu and C.-M. Pun, "Deep fusion network for splicing forgery localization," in *Proceedings of the European Conference on Computer Vision (ECCV) Workshops*, September 2018.

[19] Y. Rao, J. Ni, and H. Zhao, "Deep learning local descriptor for image splicing detection and localization," *IEEE Access*, vol. 8, pp. 25 611–25 625, 2020.

[20] B. Xiao, Y. Wei, X. Bi, W. Li, and J. Ma, "Image splicing forgery detection combining coarse to refined convolutional neural network and adaptive clustering," *Information Sciences*, vol. 511, pp. 172 – 191, 2020. [Online]. Available: http://www.sciencedirect.com/science/article/pii/S0020025519308977

[21] M. Tan and Q. V. Le, "Efficientnet: Rethinking model scaling for convolutional neural networks," 2020.

[22] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," 2015.

[23] "Casia. v2.0. http://forensics.idealtest.org/casiav2/. 2009."

[24] N. Patrick and B. Chakravarthy, "Image splicing detection technique based on illumination-reflectance model and lbp," 2020.

[25] D. M. B. P. C. S. M. V. H. R. K. L. Jaiprakash, Sahani Pooja, "Low dimensional dct and dwt feature based model for detection of image splicing and copy-move forgery," 2020.

[26] Z. H. H. Abd El-Latif Eman I, Taha Ahmed, "A passive approach for detecting image splicing based on deep learning and wavelet transform," 2020.

[27] K. L. B. J. S. Kanwal Navdeep, Girdhar Akshay, "Digital image splicing detection technique using optimal threshold based local ternary pattern," 2020.

[28] N. Y. e. a. Hussien, "Deep learning on digital image splicing detection using cfa artifacts." 2021.

[29] N. Krawetz, "A picture's worth... hacker factor solutions." 2007.

[30] B. Mahdian and S. Saic., "Using noise inconsistencies for blind image forensics. image and vision computing," 2009.

[31] P. Zhou, X. Han, V. I. Morariu, and L. S. Davis, "Learning rich features for image manipulation detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2018.

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 06:59:52 UTC from IEEE Xplore. Restrictions apply.